Tajveer Singh Dhesi

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EDUCATION

University of Chicago — *Ph.D., Computer Science* September 2025 – Present

- Advised by Prof. Nick Feamster (NOISE Lab). Research areas: web measurement, AI governance, agentic AI privacy, online safety policy.

Colgate University — *B.A., Computer Science and Mathematics (double major)* August 2021 – May 2025

- Graduated *summa cum laude*, with High Honors in Computer Science. Dean's Award with Distinction – 5 consecutive semesters. Laura Sanchis Award for Excellence in Research (2025).

LANGUAGES & TECHNOLOGIES

Programming: Python (Selenium, requests, Playwright), R, Java, JavaScript, C, SQL
ML / AI: OpenAI API, Claude Console, Hugging Face transformers, local LLM inference (Ollama, mlx.lm), privacy-aware prompt engineering, evaluation pipeline design
Systems & tooling: Linux/Unix, Docker (network namespaces), iptables, Git, LaTeX, headless Chromium, Android emulator automation, MCP (Model Context Protocol)

RESEARCH EXPERIENCE

Graduate Researcher — NOISE Lab, University of Chicago *Advisor: Prof. Nick Feamster* September 2025 – Present

Agentic AI Security & Privacy Enforcement (SIDECAR) March 2026 – Present

- Leading development of SIDECAR, a context-aware privacy enforcement layer that sits in the network path of LLM agents and blocks or redacts outbound information inappropriate for its recipient across email, web posts, and agent-to-agent messages.
- Framework-agnostic by design: enforces protections even when an agent is compromised, misconfigured, or simply over-helpful, using Docker network isolation, iptables interception, a curated type information taxonomy, recipient-specific disclosure norms, using LLM labelers.
- Pilot results: reduced information leakage by 84% on PrivacyLens (470 cases) while retaining 85% of task utility, and by 75% on AgentDAM — an auditable, fail-closed control aligned with NIST AI RMF guidance and emerging FTC and state-AG enforcement around AI privacy harms.

Behavioral Age-Assurance Auditing September 2025 – Present

- Leading a black-box audit of YouTube's production AI age-estimation system to determine which behavioral signals (search, viewing, app installs) cause self-attested adult accounts to be reclassified as minors — a deployed system whose accuracy regulators have questioned but no prior work has measured.
- Building a scalable sock-puppet measurement harness on Android emulators with accessibility-driven UI automation, replicated across multiple jurisdictions.
- Gives regulators an empirical basis to assess the necessity and proportionality of behavioral age inference under GDPR data minimisation, EU DSA Art. 28, the UK Online Safety Act, and Australia's social media minimum-age law.

Senior Thesis — Colgate University *"Measuring the Variety of Website Arrival Age Gates"* August 2024 – May 2025

- Built an automated pipeline (Selenium + headless Chromium + GPT-4o-mini classifiers) measuring site-arrival age gates across 4,985 international websites in the e-cigarette, alcohol, and gambling industries; cataloged 595 gates across 8 ccTLDs and found virtually all rely on trivial self-attestation.
- Translated findings into policy framing tied to specific jurisdictional regimes (Canada's Vaping Promotion Regulations, UK Gambling Commission LCCP, NSW alcohol delivery rules).

PUBLICATIONS

- T. S. Dhesi, N. Apthorpe. *"Measuring the Prevalence and Variety of Online Age-Gates."* ConPro 2025.
- T. S. Dhesi, A. W. Bannister, A. Manzourolajdad. *"Deep Learning of Ligand-bound RNA Tertiary Structures Diverges from Learning Unbound Ones."* bioRxiv, 2023. doi:10.1101/2023.09.13.557627.

TEACHING & SELECTED PROJECTS

Teaching Assistant, University of Chicago September 2025 – Present

- *Introduction to Computer Security*; *Security, Privacy, and Consumer Protection*; *Machine Learning for Computer Systems*; *Creating Interactive Systems with User-Centered Design*.

Teaching Assistant, Colgate University August 2023 – December 2023

- Supported 50+ students across multiple sections of Discrete Structures (COSC 290), and Data Structures & Algorithms (COSC 202), both taught in Java.